
Lecture 5: Partial Equilibrium

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1. Setting the Scene

In previous lectures we studied consumers and firms in *isolation*. We now bring them together. Studying *all* markets simultaneously is the subject of general equilibrium; here we restrict attention to a **single market** for one good while taking the prices of all other goods as fixed.

Objectives of partial equilibrium:

- understand how a market clears, and
- analyze the welfare implications of market outcomes and policy interventions.

1.1. Objects of Analysis

For a given good i we study two relations:

1. **Demand:** the (aggregate) demand function $q^d(p)$ or its inverse $p^d(q)$.
2. **Supply:** the (aggregate) supply function $q^s(p)$ or its inverse $p^s(q)$.

1.2. Setup

There are L consumers indexed by ℓ and K producers indexed by k .

- Consumer ℓ has Marshallian demand $x_{\ell,i}^*(p, I_\ell)$ for good i at total income I_ℓ .
- Producer k has supply $y_k(p, \bar{w})$ for given input prices $\bar{w}$.

The (partial-equilibrium) aggregate demand and supply for good i are

$$q^d(p) \equiv \sum_{\ell=1}^L x_{\ell,i}^*(p_i = p, \bar{p}_{-i}, I_\ell), \quad q^s(p) \equiv \sum_{k=1}^K y_k(p, \bar{w}),$$

with $\bar{p}_{-i}$ and $\bar{w}$ held fixed (this is what makes the analysis *partial*).

2. A Cobb–Douglas Worked Example

Example

Consumers. L consumers each consume two goods $i \in \{1, 2\}$ with utility $u_\ell(x) = x_1^{\alpha_1^\ell} x_2^{\alpha_2^\ell}$, $\alpha_1^\ell + \alpha_2^\ell = 1$. Standard Lagrangian gives

$$x_{\ell,i}^*(p_i = p, \bar{p}_{-i}, I_\ell) = \frac{\alpha_i^\ell I_\ell}{(\alpha_1^\ell + \alpha_2^\ell) p},$$

so the aggregate demand for good 1 is

$$q^d(p) = \frac{1}{p} \sum_{\ell=1}^L \frac{\alpha_1^\ell}{\alpha_1^\ell + \alpha_2^\ell} I_\ell.$$

Producers. Each firm k uses inputs z_A, z_B with $f_k(z) = z_A^{\beta_A^k} z_B^{\beta_B^k}$, $\beta_A^k + \beta_B^k < 1$. From Lecture 4 the firm's supply is

$$y_k(p, w) = \left(\frac{p(\beta_A^k + \beta_B^k)}{\phi_k(w_A, w_B)} \right)^{(\beta_A^k + \beta_B^k)/(1 - \beta_A^k - \beta_B^k)}.$$

Symmetric simplification. Set $\alpha_1^\ell = \alpha$, $\alpha_2^\ell = 1 - \alpha$ and $\beta_A^k = \beta_B^k = \frac{1}{4}$ for all agents. Then

$$q^d(p) = \frac{\alpha}{p} \sum_\ell I_\ell = \frac{\alpha I}{p}, \quad q^s(p) = \frac{p K}{2\phi},$$

where $I \equiv \sum_\ell I_\ell$ and ϕ is the common cost coefficient.

3. Market Equilibrium

A **partial equilibrium** in the market for good i is a price p^* and quantity q^* such that

$$q^d(p^*) = q^s(p^*) = q^*.$$

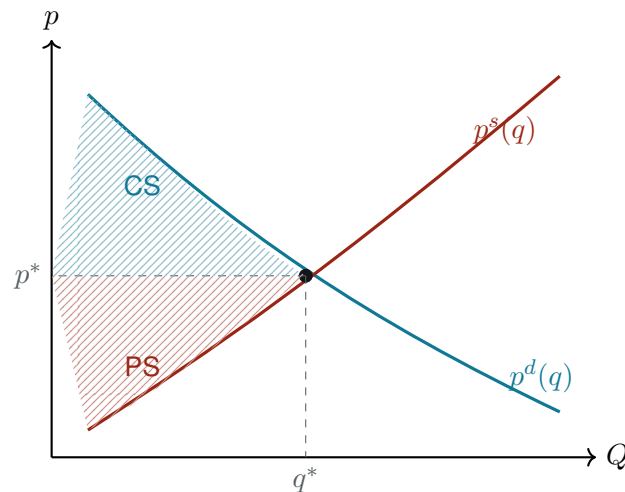
At p^* , firms maximize profit, consumers maximize utility, and the market clears.

Example

Continuing the previous example, $q^d(p^*) = q^s(p^*)$ yields

$$\frac{\alpha I}{p^*} = \frac{p^* K}{2\phi} \implies p^* = \sqrt{\frac{2\alpha I \phi}{K}}, \quad q^* = \sqrt{\frac{K \alpha I}{2\phi}}.$$

4. Welfare: Consumer and Producer Surplus



4.1. Consumer Surplus

$$CS(q) \equiv \int_0^q (p^d(s) - p^d(q)) ds = \int_0^q p^d(s) ds - p^d(q) \cdot q.$$

The area between the inverse demand curve and the equilibrium price line, from 0 to q .

4.2. Producer Surplus

$$PS(q) \equiv \int_0^q (p^s(q) - p^s(s)) ds = p^s(q) \cdot q - \int_0^q p^s(s) ds.$$

5. Interpreting Surplus

5.1. Connection to the Expenditure Function

Connecting Marshallian and Hicksian demand $x_{\ell,i}^*(p, I_\ell) = x_{\ell,i}^h(p, v_\ell(p, I_\ell))$ and using Shephard's lemma $x_{\ell,i}^h(p, \bar{u}) = \partial e_\ell(p, \bar{u}) / \partial p_i$:

$$x_{\ell,i}^*(p, I_\ell) = \frac{\partial e_\ell(p, v_\ell(p, I_\ell))}{\partial p_i}.$$

The consumer surplus can also be written as the area under demand to the right of $p^d(q)$:

$$CS(q) = \int_{p^d(q)}^\infty q^d(z) dz = \sum_{\ell=1}^L \int_{p^d(q)}^\infty x_{\ell,i}^*(z, \bar{p}_{-i}, I_\ell) dz.$$

Quasi-separable case. If $e_\ell(p, \bar{u}) = f_\ell(p) + g_\ell(\bar{u})$, then $\partial e_\ell / \partial p_i$ does *not* depend on utility, and

$$CS(q) = \sum_{\ell} [e_\ell(\infty, \bar{p}_{-i}, \bar{u}) - e_\ell(p^d(q), \bar{p}_{-i}, \bar{u})].$$

Interpretation: CS measures the amount of money consumers would need to be compensated to leave the market for good i . It quantifies, in dollars, the benefit of being able to participate in the market.

5.2. Quasi-Linear Utility

A common (and clean) assumption is quasi-linear utility,

$$u(x, m) = \mu(x) + m, \quad \mu'(x) > 0, \mu''(x) < 0, \mu'(0) = \infty, \lim_{x \rightarrow \infty} \mu'(x) = 0,$$

where m is money spent on *everything else*. The consumer solves

$$\max_{x, m \geq 0} \mu(x) + m \quad \text{s.t.} \quad px + m \leq y.$$

First-order conditions. Interior solution gives

$$\mu'(x) = \lambda p, \quad 1 = \lambda \implies \mu'(x) = p \implies x^*(p) = (\mu')^{-1}(p).$$

Key feature: no income effect on x . The choice of x depends only on p , not on y .

Budget balance gives $m^*(p, y) = y - p x^*(p)$, and the indirect utility is

$$v(p, y) = \mu(x^*(p)) + y - p x^*(p).$$

Using the identity $v(p, e(p, \bar{u})) = \bar{u}$,

$$e(p, \bar{u}) = \bar{u} - \mu(x^*(p)) + p x^*(p).$$

The expenditure function is *additively separable* in p and $\bar{u}$. This is exactly the condition under which the Marshallian and Hicksian CS coincide — so the area under the demand curve really is the right welfare measure.

5.3. Producer Surplus via Profit

By Hotelling's Lemma $\partial \pi_k / \partial p = y_k(p)$, so

$$\text{PS}(q) = \int_0^{p^s(q)} q^s(z) dz = \sum_{k=1}^K \int_0^{p^s(q)} y_k(z) dz = \sum_{k=1}^K \int_0^{p^s(q)} \frac{\partial \pi_k(z)}{\partial p} dz = \sum_{k=1}^K [\pi_k(p^s(q)) - \pi_k(0)].$$

Since $\pi_k(0) = 0$ (no production if the firm doesn't operate), producer surplus is the sum of firms' profits: the dollar benefit producers derive from the existence of the market.

6. Application: Specific Taxes

6.1. Setup

A specific (per-unit) tax t is levied on suppliers. Consumers pay $p + t$, producers receive p . The equilibrium price $p^*(t)$ is determined by

$$q^d(p^*(t) + t) = q^s(p^*(t)) \equiv q^*(t).$$

6.2. Effect on the Equilibrium Price

Differentiating implicitly,

$$q_p^d(p^*(t) + t) (p_t^*(t) + 1) = q_p^s(p^*(t)) p_t^*(t),$$

where $q_p^d = \partial q^d / \partial p < 0$ and $q_p^s > 0$. Solving for $p_t^*(t)$,

$$p_t^*(t) = \frac{q_p^d(p^*(t) + t)}{q_p^s(p^*(t)) - q_p^d(p^*(t) + t)} = \frac{1}{\frac{q_p^s(p^*(t))}{q_p^d(p^*(t) + t)} - 1} < 0.$$

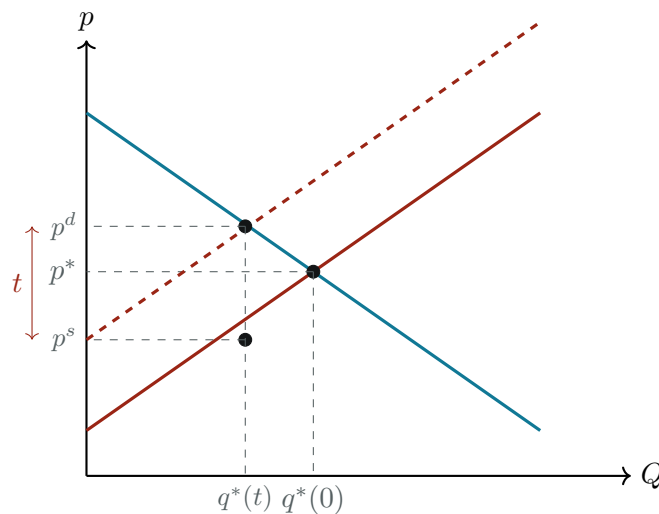
Two takeaways.

1. $p_t^*(t) < 0$: a tax lowers the price received by producers.
2. The *magnitude* depends on the ratio of supply to demand slopes. When supply is much more elastic than demand, the producer price falls less and the consumer price rises more (the burden shifts onto consumers).

Price paid by consumers. $\partial(p^*(t) + t) / \partial t = p_t^*(t) + 1 > 0$, so consumers pay strictly more after the tax is introduced.

6.3. Effect on the Equilibrium Quantity

$$q_t^*(t) = q_p^s(p^*(t)) p_t^*(t) = \underbrace{q_p^s(p^*(t))}_{>0} \underbrace{p_t^*(t)}_{<0} < 0.$$



7. Welfare Cost of a Tax

The total welfare under tax t is

$$ET(t) = CS(t) + PS(t) + R(t),$$

where $R(t) = q^*(t) \cdot t$ is government revenue.

Expanding the integrals,

$$CS(t) = \int_0^{q^*(t)} (p^d(s) - (p^*(t) + t)) ds, \quad PS(t) = \int_0^{q^*(t)} (p^*(t) - p^s(s)) ds, \quad R(t) = t q^*(t).$$

Adding everything (the $p^*(t)$ and t terms cancel),

$$\text{ET}(t) = \int_0^{q^*(t)} (p^d(s) - p^s(s)) ds.$$

7.1. Deadweight Loss

The reduction in total surplus relative to the laissez-faire benchmark $\text{ET}(0)$ is the **deadweight loss**:

$$\text{DWL}(t) \equiv \text{ET}(0) - \text{ET}(t) = \int_{q^*(t)}^{q^*(0)} (p^d(s) - p^s(s)) ds \geq 0.$$

Differentiating ET with respect to t ,

$$\text{ET}'(t) = (p^d(q^*(t)) - p^s(q^*(t))) q_t^*(t) = t q_t^*(t) \leq 0,$$

since at the tax equilibrium $p^d - p^s = t$ and $q_t^* \leq 0$. Welfare is strictly decreasing in the tax (away from $t = 0$).

8. Optimal Taxation

Suppose tax revenue generates social benefits not captured by private agents:

$$G(t q^*(t)) = K t q^*(t), \quad K > 0.$$

A social planner maximizes

$$\max_t K R(t) + \text{ET}(t) = K t q^*(t) + \int_0^{q^*(t)} (p^d(s) - p^s(s)) ds.$$

First-order condition:

$$K q^*(t) + K t q_t^*(t) + \text{ET}'(t) = 0 \implies K q^*(t) + K t q_t^*(t) + t q_t^*(t) = 0.$$

Solving for the optimal tax t^* ,

$$t^* = \frac{K}{K+1} \frac{q^*(t^*)}{-q_t^*(t^*)}.$$

The optimal tax is increasing in the social value K of revenue and decreasing in $|q_t^*|$: the more sensitive equilibrium quantity is to the tax, the smaller the optimal tax. This is the inverse-elasticity rule.

Reference: Jehle & Reny, Advanced Microeconomic Theory, Chapter 4.